

OS-UAED: OPEN-SET UNIFIED AUDIO EVENT DETECTION WITH SPEAKER-AWARE NONVERBAL EVENTS

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ABSTRACT

Unified audio event detection (UAED) aims to jointly capture the activities of different speakers and general sound events. However, it is typically limited to predefined event classes and provides limited support for speaker-attributed nonverbal events. We propose OS-UAED, an open-set UAED framework with semantic-guided residual transfer. Built upon a set of closed-set base classes, the proposed framework enables adaptive transfer to unseen sound classes by transforming the text/audio-based novel queries into task-adapted representations. OS-UAED further combines speaker identity with nonverbal semantics to generate speaker-aware queries for vocalizations such as laughter and coughing. Experiments demonstrate effective generalization to novel sound events, improved speaker-aware nonverbal detection, and competitive downstream sound event detection (SED) performance on DESED. Ablation results confirm the importance of the proposed query transfer mechanism.

Index Terms— Unified audio event detection, Open-set sound event detection, Speaker diarization, Nonverbal sound

1. INTRODUCTION

Real-world acoustic scenes contain heterogeneous and overlapping activities, including speech, environmental sounds, and nonverbal vocalizations such as laughter and coughing. Unified audio event detection (UAED) jointly models these activities within a single framework [1], and recent extensions further incorporate speaker-associated nonverbal events [2]. However, existing UAED systems remain largely closed-set, restricting sound-event detection to predefined categories and limiting their ability to handle unseen acoustic events.

Pretrained audio-language models such as CLAP [3] provide transferable semantic representations for unseen sound concepts. Recent open-vocabulary sound event detection (SED) approaches, including MGA-CLAP [4], FLAM [5], FlexSED[6] and DASM [7], exploit such representations to detect events beyond the training taxonomy. Nevertheless, these methods primarily focus on sound-event detection rather than unified modeling of speakers, environmental events, and speaker-attributed nonverbal activities. Moreover,

directly using CLAP embeddings as detector queries may be suboptimal, because the semantic space learned for audio-text alignment is not necessarily aligned with the task-adapted query space required for frame-level detection.

To address these limitations, we propose OS-UAED, an open-set unified audio event detection framework with semantic-guided residual transfer. For each unseen event, a novel query based on either text descriptions or audios is accepted as auxiliary input. CLAP-embedding-based similarities then identify the most relevant base-class semantic anchors, whose learned representations are utilized to construct the task-adapted query. OS-UAED further combines speaker identity with nonverbal semantics to generate speaker-aware queries for vocalizations such as laughter and coughing. All query types share a common audio representation and are jointly processed by a cross-modal decoder for frame-level localization. Our contributions are threefold:

1. We extend UAED from closed-set to open-set sound event detection while retaining unified modeling of heterogeneous acoustic activities.
2. We introduce semantic-guided residual transfer, which constructs task-adapted novel-event queries by transferring learned representations from semantically related base classes.
3. We incorporate speaker-aware nonverbal event detection and evaluate OS-UAED on open-set detection, unified audio event detection, and downstream SED.

2. OPEN-SET UNIFIED AUDIO EVENT DETECTION

2.1. Overall architecture and query generation

OS-UAED formulates heterogeneous acoustic activities as query-conditioned frame-level detection targets. As shown in Fig. 1, an input mixture $\mathbf{X}$ is processed by WavLM [8], BEATs [9], and CNN-based branches to capture speaker, sound-event, and local acoustic information, respectively. Their features are temporally aligned via linear interpolation and then concatenated, projected (for fusion), and encoded as

$$\mathbf{H} = E_{\text{enc}}(F_{\text{fuse}}[\mathbf{H}^{\text{WavLM}}; \mathbf{H}^{\text{BEATs}}; \mathbf{H}^{\text{CNN}}]). \quad (1)$$

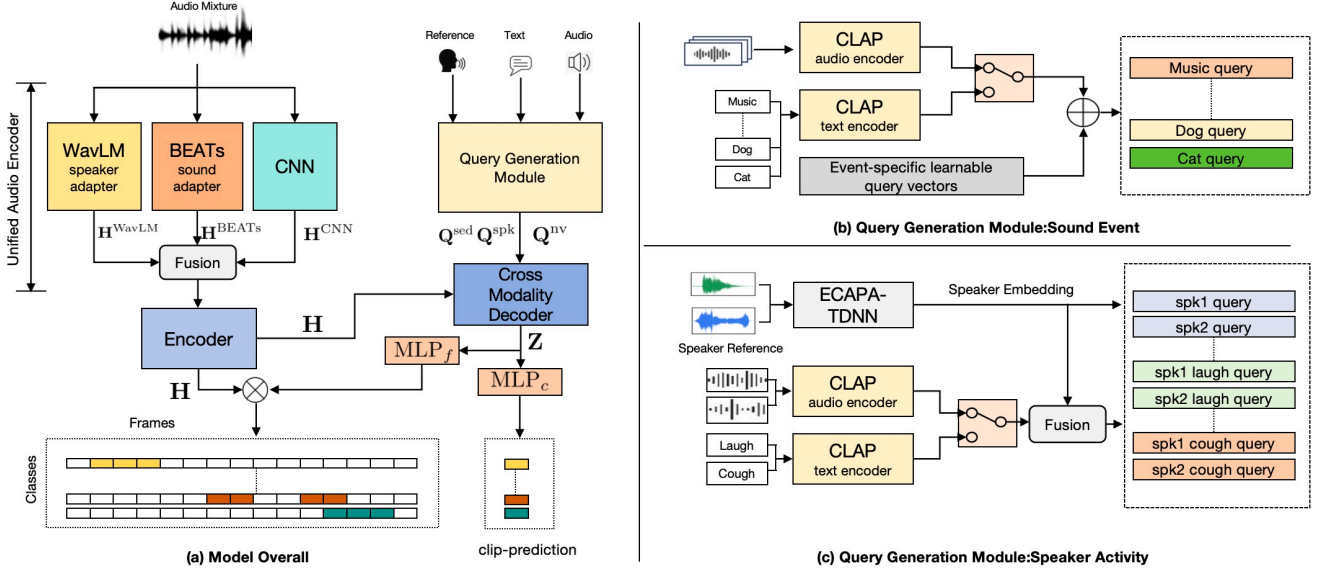


Fig. 1: Overview of OS-UAED: (a) overall architecture, (b) sound-event query generation, and (c) speaker and speaker-aware nonverbal query generation. “Fusion” denotes feature-wise concatenation followed by a linear projection to the model hidden dimension.

Sound-event, speaker, and speaker-aware nonverbal targets are represented by learnable queries $\mathbf{Q}^{\text{sed}}$, $\mathbf{Q}^{\text{spk}}$, and $\mathbf{Q}^{\text{nv}}$, which interact with the shared audio representation through a cross-modal decoder $D(\cdot)$:

$$\mathbf{Z} = D([\mathbf{Q}^{\text{sed}}; \mathbf{Q}^{\text{spk}}; \mathbf{Q}^{\text{nv}}], \mathbf{H}), \quad (2)$$

where $\mathbf{Z} \in \mathbb{R}^{N_q \times d}$ denotes the refined query representations. N_q is the number of queries, and d is the feature dimension.

We employ separate prediction heads for temporal localization and clip-level presence. The refined queries are first transformed by a frame prediction MLP,

$$\mathbf{Z}_f = \text{MLP}_f(\mathbf{Z}), \quad (3)$$

and frame-level probabilities are obtained by scaled dot-product scoring with the encoder features,

$$\mathbf{Y}_f = \sigma \left(\frac{\mathbf{Z}_f \mathbf{H}^\top}{\sqrt{d}} \right). \quad (4)$$

In parallel, a clip prediction MLP directly produces clip-level probabilities,

$$\mathbf{Y}_c = \sigma(\text{MLP}_c(\mathbf{Z})). \quad (5)$$

Thus, frame-level and clip-level supervision jointly optimize the shared query representations.

For each base sound-event class c_i^B , CLAP encodes its text or audio reference r_i^B , which is projected into the query space,

$$\mathbf{p}_i^B = P_{\text{clap}}(f_{\text{clap}}(r_i^B)). \quad (6)$$

Each base class additionally maintains a learnable task-specific vector $\mathbf{l}_i^B$. The final base query is

$$\mathbf{q}_i^B = g\mathbf{p}_i^B + (1 - g)\mathbf{l}_i^B, \quad (7)$$

where γ is a learnable scalar parameter and $g = \sigma(\gamma)$ is the corresponding global fusion coefficient.

For speaker-related targets, a pretrained ECAPA-TDNN model [10] extracts a speaker embedding $\mathbf{e}_s^{\text{spk}}$ from the reference speech. A speaker-aware nonverbal query is then constructed by combining it with the CLAP representation $\mathbf{e}_v^{\text{nv}}$:

$$\mathbf{q}_{s,v}^{\text{nv}} = \mathbf{W}_{\text{nv}}[\mathbf{e}_s^{\text{spk}}; \mathbf{e}_v^{\text{nv}}] + \mathbf{b}_{\text{nv}}, \quad (8)$$

where $\mathbf{W}_{\text{nv}}$ and $\mathbf{b}_{\text{nv}}$ are the learnable weight matrix and bias, respectively. This explicitly represents speaker-event pairs such as speaker-specific laughter and coughing.

2.2. Semantic-guided residual transfer

Novel classes are unseen during training and therefore lack class-specific task adaptation. Given a novel reference r_n^N , CLAP produces

$$\mathbf{e}_n^N = f_{\text{clap}}(\mathbf{r}_n), \quad (9)$$

Its semantic similarity to each base class is computed in the original CLAP space,

$$s_{n,i} = \cos(\mathbf{e}_n^N, \mathbf{e}_i^B), \quad i = 1, \dots, N_B. \quad (10)$$

We retain the Top- K semantic anchors $\mathcal{K}_n$ and compute temperature-normalized weights,

$$\alpha_{n,i} = \frac{\exp(s_{n,i})}{\sum_{j \in \mathcal{K}_n} \exp(s_{n,j})}, \quad i \in \mathcal{K}_n, \quad (11)$$

Table 1: Comparison of open-set and unified audio event detection performance. PSDS is reported separately for base, novel, and all sound-event classes. UAED and SED performance is evaluated using IB-F1 and SB-F1, while speaker diarization is evaluated using DER.

Model	Query	PSDS			UAED Results		SED Results		SD Results DER (MS/FA/SC) [↓]
		base [↑]	novel [↑]	all [↑]	IB-F1 [↑]	SB-F1 [↑]	IB-F1 [↑]	SB-F1 [↑]	
MGA-CLAP	text	0.259	0.369	0.276	-	-	-	-	-
FLAM	text	0.350	0.466	0.369	-	-	-	-	-
DASM	text	0.572	0.556	0.557	-	-	-	-	-
DASM	audio	0.636	0.562	0.569	-	-	-	-	-
T-UAED	learnable	0.559	-	-	0.432	0.645	0.403	0.623	5.76 (1.38/0.90/3.49)
OS-UAED	text	0.724	0.611	0.706	0.481	0.712	0.491	0.707	8.45 (3.25/2.35/2.25)
OS-UAED	audio	0.732	0.691	0.725	0.508	0.715	0.519	0.708	8.91 (3.17/2.29/3.45)

Rather than transferring complete base queries, we extract their task-adaptation residuals,

$$\mathbf{r}_i^B = \mathbf{q}_i^B - \mathbf{p}_i^B, \quad (12)$$

and aggregate them according to semantic relevance,

$$\hat{\mathbf{r}}_n = \sum_{i \in \mathcal{K}_n} \alpha_{n,i} \mathbf{r}_i^B. \quad (13)$$

The novel query is then constructed as

$$\mathbf{q}_n^N = P_{\text{clap}}(\mathbf{e}_n^N) + \beta \hat{\mathbf{r}}_n. \quad (14)$$

Here, β controls the transferred task-adaptation strength. This formulation preserves the semantic identity of the unseen class while transferring detection-specific knowledge from related base classes. Residual transfer is suppressed when no sufficiently related base class is found.

2.3. Training objective

OS-UAED is optimized using a frame-level detection loss together with gate and semantic regularization:

$$\mathcal{L} = \mathcal{L}_{\text{frame}} + \lambda_c \mathcal{L}_{\text{clip}} + \lambda_g \mathcal{L}_{\text{gate}} + \lambda_s \mathcal{L}_{\text{sem}}, \quad (15)$$

where $\mathcal{L}_{\text{frame}}$ and $\mathcal{L}_{\text{clip}}$ are binary cross-entropy losses for frame and clip level predictions, respectively.

To stabilize the fusion between the CLAP semantic representation and the event-specific learnable representation, we weakly regularize the global fusion gate toward a CLAP-biased prior:

$$\mathcal{L}_{\text{gate}} = (g - g_0)^2, \quad (16)$$

where g_0 denotes the predefined fusion prior. This regularization discourages the fusion from collapsing to either the semantic or task-adapted branch while retaining the flexibility to optimize g through the detection objective.

Furthermore, dimensionality projection from the original CLAP space to the decoder query space may distort the semantic relationships among event classes. We therefore introduce a semantic preservation loss that aligns pairwise cosine similarities before and after projection:

$$\mathcal{L}_{\text{sem}} = \frac{2}{N_B(N_B-1)} \sum_{i < j} [\cos(\tilde{\mathbf{e}}_i^B, \tilde{\mathbf{e}}_j^B) - \cos(\mathbf{e}_i^B, \mathbf{e}_j^B)]^2, \quad (17)$$

where $\mathbf{e}_i^B$ and $\tilde{\mathbf{e}}_i^B = P_{\text{clap}}(\mathbf{e}_i^B)$ denote the original and projected CLAP representations of the base class i , respectively. This objective preserves the relative semantic structure of the pretrained CLAP space while adapting the representations to frame-level detection.

3. EXPERIMENTAL SETUP

Datasets. We construct simulated mixtures from three acoustic sources. Sound events are selected from FSD50K [11] using the clean single-label clips identified by FSD50K-Solo [12], comprising 43 base classes and 8 novel classes¹. Novel classes are completely excluded from training and introduced only at inference through text descriptions or external audio exemplars. Speech is sampled from LibriSpeech [13] with three speakers per mixture, while speaker-associated laughter and coughing are synthesized following the simulation pipeline of SA-UAED [2]. OS-UAED is further fine-tuned and evaluated on DESED[14] for downstream SED.

Evaluation Metrics. We use the polyphonic sound detection score (PSDS) [15] as the primary SED metric. Following prior work [16, 17], the class variance penalty is omitted, as it was designed for more balanced datasets. For open-set evaluation, PSDS is reported separately on base, novel, and all classes. We additionally report intersection-based F1 scores (IB-F1) [15] and segment-based F1 scores (SB-F1) [18] for UAED/SED, and the diarization error rate (DER) for speaker

¹Provide a Demo page to introduce the detailed classes and show audio examples.

Table 2: Speaker-Aware Nonverbal Event Detection results with different query modalities. IB-F1 and SB-F1 are reported for laughter and coughing.

Model	Query	laugh		Cough	
		IB-F1 $\uparrow$	SB-F1 $\uparrow$	IB-F1 $\uparrow$	SB-F1 $\uparrow$
SA-UAED	-	0.331	0.687	0.359	0.708
OS-UAED	audio	0.450	0.745	0.434	0.733
	text	0.413	0.722	0.401	0.714
	audio/text	0.437	0.733	0.427	0.713

Table 3: Effect of the residual-transfer weight β on novel-class PSDS. $\beta = 0$ denotes CLAP-only novel queries without residual transfer.

query	β	0	0.25	0.5	0.75	1
audio		0.009	0.057	0.486	0.691	0.647
text		0.008	0.038	0.417	0.611	0.426

diarization. PSDS1 and PSDS2 are used for downstream evaluation on DESED.

Baselines. Open-set SED baselines include MGA-CLAP [4], FLAM [5], and DASM [7]. T-UAED [1] serves as the closed-set unified-detection baseline, and SA-UAED [2] is used for speaker-aware nonverbal detection. For downstream SED on DESED, we further compare with representative SED systems, including AST-SED [19], ATST-SED [20], CRNN-BEATs, PaSST-SED [21], and MFDCConv-BEATs [22].

Implementation details. The audio encoder is a 3-layer, 8-head Conformer encoder, and the cross-modal decoder is a 6-layer, 8-head Transformer decoder. We adopt MGA-CLAP[4] for audio-text representations. OS-UAED contains 423.04M parameters, of which only 16.99M are trainable, as the pre-trained models are frozen. We use $K = 2$, $\tau = 0.07$, and residual weight $\beta = 0.75$. We set $\lambda_c = 0.5$ $\lambda_g = \lambda_s = 0.01$ and use a mildly CLAP-biased gate prior $g_0 = 0.65$.

4. RESULTS

4.1. Open-Set and unified audio event detection

Table 1 compares OS-UAED with open-set SED baselines and the closed-set UAED baselines. With text queries, OS-UAED achieves a novel-class PSDS of 0.611, exceeding DASM (0.556), FLAM (0.466), and MGA-CLAP (0.369). Audio queries further improve novel PSDS to 0.691, compared with 0.562 for DASM. OS-UAED also improves base/all-class PSDS and UAED/SED F1 scores over T-UAED, although DER increases under the current joint detection setting.

4.2. Speaker-aware nonverbal event detection

Table 2 shows consistent gains over SA-UAED. With audio queries, laughter IB-F1/SB-F1 improves from 0.331/0.687 to

Table 4: Downstream SED results on DESED after fine-tuning. PSDS1 and PSDS2 are reported for different baselines and OS-UAED query modalities.

Model	PSDS1 $\uparrow$	PSDS2 $\uparrow$
AST-SED [19]	0.514	-
ATST-SED (BEATs freeze) [20]	0.501	0.755
ATST-SED (ATST finetune) [20]	0.583	0.810
CRNN-BEATs	0.500	0.762
PaSST-SED [21]	0.555	0.791
MFDCConv-BEATs [22]	0.552	0.794
OS-UAED-SED (audio)	0.554	0.738
OS-UAED-SED (text)	0.551	0.762

0.450/0.745, while coughing improves from 0.359/0.708 to 0.434/0.733, demonstrating the benefit of explicitly conditioning nonverbal queries on speaker identity.

4.3. Residual-transfer ablation

Table 3 studies the transfer weight β . CLAP-only novel queries ($\beta = 0$) obtain only 0.009/0.008 PSDS for audio/text queries. Performance increases substantially with residual transfer and peaks at $\beta = 0.75$, reaching 0.691 and 0.611, respectively. Full residual transfer ($\beta = 1$) degrades performance, indicating that moderate transfer better balances semantic preservation and task-specific adaptation.

4.4. Downstream sound event detection

Table 4 evaluates whether the unified representations learned by OS-UAED can transfer to conventional closed-set SED after fine-tuning on DESED. Despite being designed for unified and open-set detection rather than specialized SED, OS-UAED achieves competitive performance: the audio-query model obtains a PSDS1 of 0.554, comparable to PaSST-SED (0.555) and MFDCConv-BEATs (0.552), while the text-query model reaches a PSDS2 of 0.762, matching CRNN-BEATs. These results indicate that the proposed unified representation retains strong transferability to standard SED without requiring a task-specific redesign.

5. CONCLUSION

We presented OS-UAED, an open-set UAED framework that transfers task-adaptation residuals from semantically related base classes to construct novel-event queries while supporting speaker-aware nonverbal detection. Experiments demonstrate effective novel-class generalization, improved unified detection, and strong transferability to downstream SED. Future work will scale OS-UAED to larger sound taxonomies, richer speaker-attributed nonverbal events, and more diverse real-world acoustic scenes, toward broader coverage of everyday acoustic events.

6. ACKNOWLEDGMENT

Generative AI is used for ...

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